Observation of Impurity Band Transport in Al-rich n-AlGaN films using Low-temperature MOCVD

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UWBG AlGaN is indispensable for next-generation electronic and optoelectronic devices, where achieving high free carrier concentration (n_s) is critical for reducing resistance, despite challenges from defects, self-compensation, and DX centers in Si-doped layers [1]. Generally, the n-AlGaN thin films are deposited at high temperatures (HT) (≥ 1100 °C) using metal-organic-chemical vapor deposition (MOCVD) technique [2,4], which causes severe group-III vacancy formations, thus limiting the conductivity [1-4]. This study demonstrates that low-temperature (LT) deposition of UWBG n-AlGaN films offers advantages over high-temperature (HT) growth, particularly in enhancing conductivity. A high n_s of 5.2×10^{19} to 8.52×10^{18} cm^{-3} and ultra-low resistivity (ρ) of 2.4×10^{-3} to 2.4×10^{-2} Ω.cm was obtained for LT n-$Al_xGa_{1-x}N$ (x_{Al}: 0.45 to 0.84) films compared to the HT n-AlGaN film, which showed a maximum charge of 1.32×10^{19} cm^{-3} and ρ of 1.5×10^{-2} Ω.cm for x_{Al}: 0.46. A temperature-independent n_s and mobility (μ) profile was obtained for Al-content up to 0.84, indicating the LT-MOCVD process as a promising route for obtaining highly conductive UWBG n-AlGaN films. The LT-MOCVD technique reduced the overall compensation ratio in n-AlGaN films compared to the conventional high-temperature (> 1100 °C) (HT) MOCVD technique, enhancing doping efficiency.

Si-doped $Al_xGa_{1-x}N$ (~70 nm) layers were deposited on AlN/Sapphire templates using the MOCVD technique at two different temperatures, 1235 °C (HT) and 985 °C (LT). Trimethylaluminum (TMAl), triethylgallium (TEGa), and ammonia (NH_3) were used as metal-organic and hydride precursors, respectively. 100 ppm Silane gas (diluted in N_2) was used as a Si dopant source. Al-content was varied by modifying the TEGa flow rate and keeping the TMAl and V/III ratio the same. At first, three different Al content $Al_xGa_{1-x}N$ films ($x_{Al,HT}$: ~0.46, ~0.65, ~0.79; and $x_{Al,LT}$: ~0.45, ~0.64, ~0.84) with different Si doping concentrations (1-10×10^{19} cm^{-3}) were deposited using HT and LT MOCVD processes. After optimizing the Si concentration to maximize n_s for both HT and LT processes, n-AlGaN films with $x_{Al,HT}$ ~ 0.86 and $x_{Al,LT}$ ~ 0.74 were deposited to evaluate the deposition-temperature-dependent resistivity (ρ) across different Al compositions. The Al-content and Si concentration were measured using XRD and SIMS measurement techniques. Electrical properties of n-$Al_xGa_{1-x}N$ films were characterized by Hall measurements using the Van der Pauw geometry at room temperature and in the 150–450 K range.

RT Hall measurement demonstrated that n-AlGaN films deposited at HT showed $n_s < 2\times10^{19}$ cm^{-3} (Fig. 1a) for all $x_{Al,HT}$, compared to n-AlGaN films deposited at LT, which showed a maximum n_s of 5.2×10^{19} cm^{-3} (Fig. 1b). Overall compensation ratio was much higher for $x_{Al,HT}$ (Fig. 2a) compared to $x_{Al,LT}$ (Fig. 2b), which indicated that compensating mechanisms such as the defects, self-compensations and DX formation were relatively lower in LT deposition of n-AlGaN. A temperature-dependent Hall measurement of the LT n-AlGaN showed pure impurity band transport with temperature-independent n_s and μ profile (Fig. 3a and Fig. 3b). The resistivity (Fig. 4a) and the E_A (Fig. 4b) of the LT n-AlGaN films were compared with the state-of-the-art n-AlGaN films deposited using MOCVD from literature, and it was found that this work shows ultra-low resistivity and the lowest E_A. The E_A was calculated from the slope of the Arrhenius plot of ln(n_s) vs 1/T profile, which demonstrated E_A < 15 meV for all the $x_{Al,LT,}$ including x_{Al} ~0.74 (not shown here). Thus, an n_s of 8.52×10^{18} cm^{-3} and ultra-low resistivity of 2.4×10^{-2} Ω.cm was obtained for $x_{Al,LT}$ ~ 0.84, which is among the state-of-the-art reported data. As impurity band transport dominates in the degenerately doped regime, precise control of MOCVD deposition conditions can enable very high n_s in n-$Al_xGa_{1-x}N$ (x_{Al} > 0.8) films. In this case, conduction is determined by the density of delocalized impurity states, rather than the donor activation energy relative to the conduction band. Electron transport is dominated by hopping between impurity donor states instead of thermal excitation to the conduction band. Consequently, both carrier concentration and mobility exhibit weak temperature dependence when transport occurs through the impurity band. Moreover, this temperature-independent conduction behavior is advantageous for the efficient operation of devices in extreme environments, which are suitable for UWBG devices.

[1] S. Washiyama et al, *J. Appl. Phys.* 127, 105702 (2020), [2] P. Bagheri *et al.*, *J. Appl. Phys.*,132, 18, (2022), [3] W. A. Doolittle et al, *Appl. Phys. Lett.* 123, 070501 (2023), [4] S. Mukhopadhyay et al, *Appl. Phys. Lett.* 125, 222102 (2024).

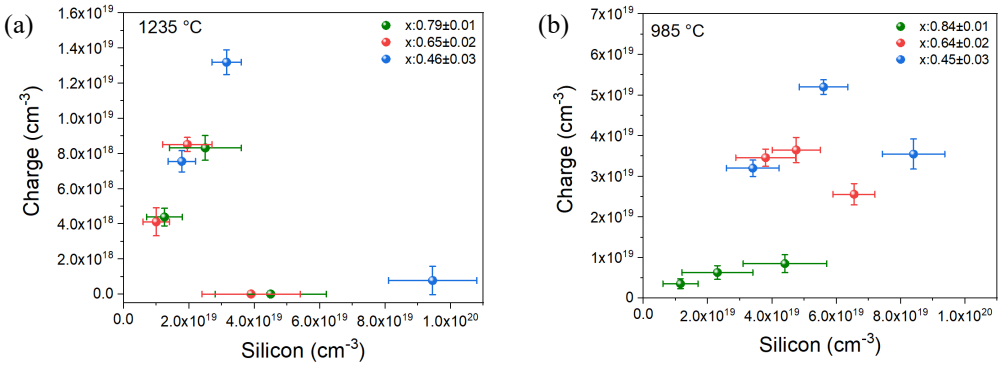

Figure 1. Comparison of charge and Si doping concentration for different n-$Al_xGa_{1-x}N$ for (a) 1235 °C and (b) 985 °C deposition temperature

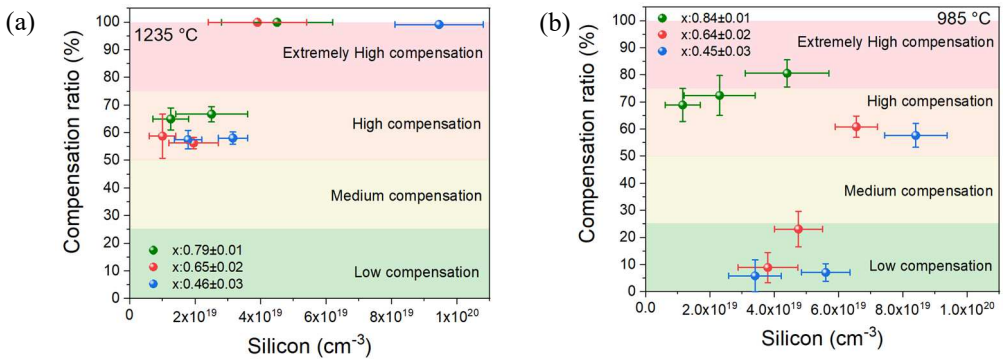

Figure 2. Comparison of compensation ratio and Si doping concentration for different n-$Al_xGa_{1-x}N$ at (a) 1235 °C and (b) 985 °C deposition temperature

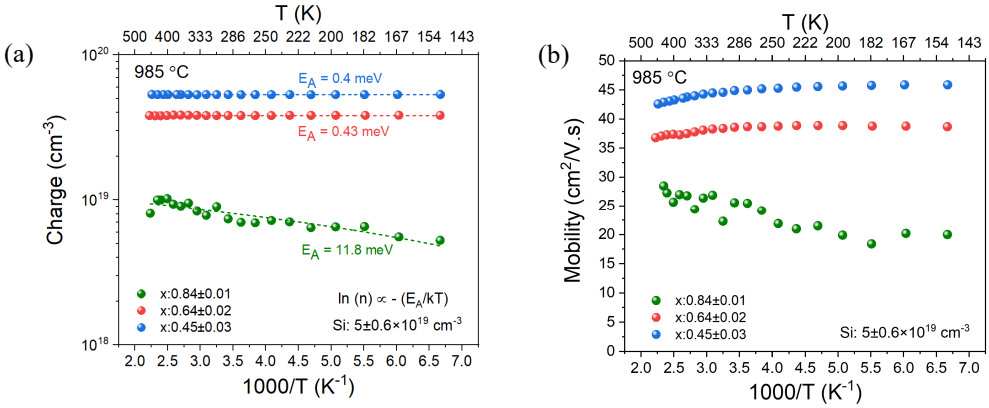

Figure 3. Temperature-dependent Hall measurement of different LT n-$Al_xGa_{1-x}N$ for obtaining (a) charge (b) mobility vs 1000/T

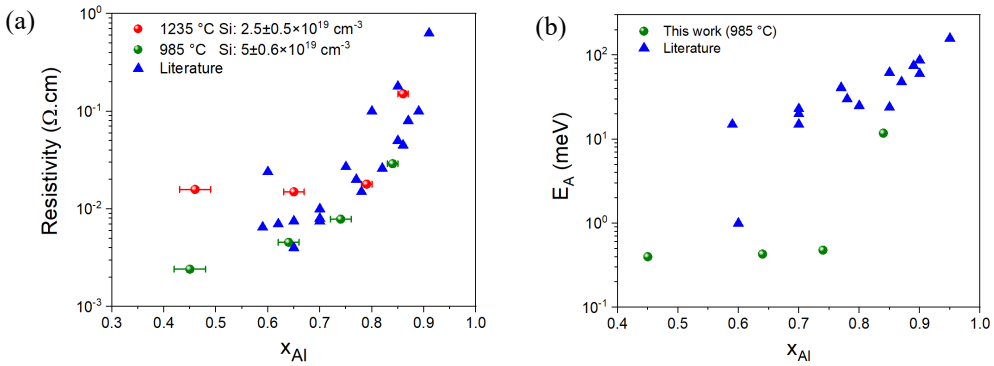

Figure 4. Benchmarking of (a) state-of-the-art resistivity and (b) activation energy for Al-rich n-$Al_xGa_{1-x}N$ films